3.2 Linear Momentum & Conservation

Question Paper

Course	CIE A Level Physics
Section	3. Dynamics
Topic	3.2 Linear Momentum & Conservation
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

What is the principle of conservation of momentum?

A. force is equal to the rate of change of momentum

B. momentum is the product of mass and velocity

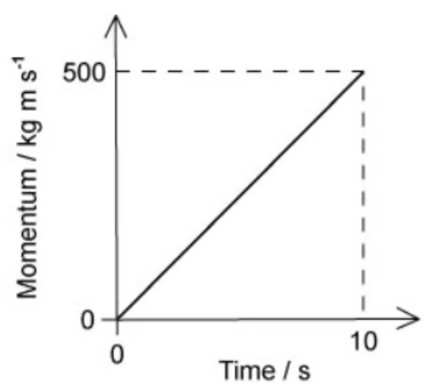
C. the total momentum of two bodies after collision is equal to their total momentum before collision

D. the total momentum of a system remains constant provided no external force acts on it

[1 mark]

Question 2

The graph shows how the momentum of a motorcycle changes with time.



What is the resultant force on the motorcycle?

A. 50 N

B. 500 N

C. 2500 N

D. 5000 N

[1 mark]

Question 3

Two bodies travelling in a straight line collide in a perfectly elastic collision. Which of the following statements must be correct?

A. the initial speed of one body will be the same as the final speed of the other body

B. the relative speed of approach between the two bodies equals their relative speed of separation

C. the total momentum is conserved but the total kinetic energy will be reduced

D. one of the bodies will be stationary at one instant

[1 mark]

Question 4

Which row correctly states whether momentum and kinetic energy are conserved in an inelastic collision in which there are no external forces?

	Momentum	**Kinetic energy**
A	conserved	conserved
B	not conserved	conserved
C	conserved	not conserved
D	not conserved	not conserved

[1 mark]

Question 5

Which quantity has the same base units as momentum?

A. density × volume × velocity

B. density × energy

C. pressure × area

D. weight ÷ area

[1 mark]

Question 6

Which of the following is a statement of the principle of conservation of momentum?

A. in an elastic collision momentum is constant

B. momentum is the product of mass and velocity

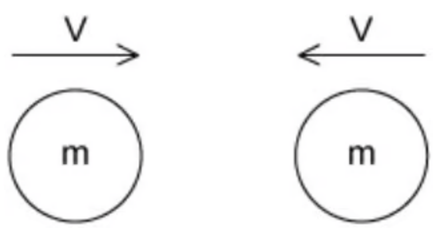
C. the force acting on a body is proportional to its rate of change of momentum

D. the momentum of an isolated system is constant

[1 mark]

Question 7

Two similar spheres, each of mass *m* and travelling with speed *v*, are moving towards each other



The spheres have a head-on elastic collision

Which statement is correct?

A. the spheres stick together on impact

B. the total kinetic energy after impact is mv^2

C. the total kinetic energy before impact is zero

D. the total momentum before impact is 2*mv*

[1 mark]

Question 8

Trolley X, moving along a horizontal frictionless track, collides with a stationary trolley Y. The two trolleys become attached and move off together.

Which statement about this interaction is correct?

A. some of the kinetic energy of trolley X is changed to momentum in the collision

B. some of the momentum of trolley X is changed to kinetic energy in the collision

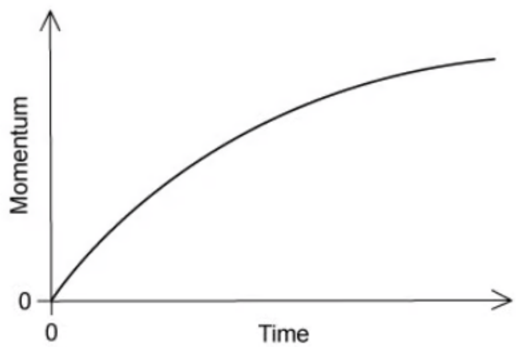
C. trolley X loses some of its momentum as heat in the collision

D. trolley X shares its momentum with trolley Y but some of its kinetic energy is lost

[1 mark]

Question 9

A car accelerates from rest. The graph shows the momentum of the car plotted against time.



What is the meaning of the gradient of the graph at a particular time?

A. the velocity of the car

B. the resultant force on the car

C. the kinetic energy of the car

D. the rate of change of kinetic energy of the car

[1 mark]

Question 10

Which statement about a ball that strikes a tennis racket and rebounds is **always** correct?

A. total kinetic energy of the ball is conserved

B. total kinetic energy of the system is conserved

C. total momentum of the ball is conserved

D. total momentum of the system is conserved

[1 mark]

Model Answers: Easy

1

The correct answer is **D** because:

- The principle of conservation of momentum states that:
  - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system
- In other words, in a **closed** system, the total momentum of two bodies before a collision is equal to their total momentum after the collision

A & B are incorrect because these are definitions of force and momentum

C is incorrect because the statement is true given there are no external forces acting, it needs clarification that this is only true in a closed or isolated system

2

The correct answer is **A** because:

- Force is equal to the rate of change of momentum
- This is equal to the gradient of a momentum-time graph
  - Force=$Force = \frac{\Delta y}{\Delta x} = \frac{500}{10} = 50\ N$

3

The correct answer is **B** because:

- A perfectly elastic collision is defined as a collision in which the initial kinetic energy equals the final kinetic energy
- In other words, the relative speed of approach equals the relative speed of separation of the two bodies because kinetic energy and speed is related by
  - $E_k = \frac{1}{2}mv^2$

A & D are incorrect because these are possible elastic collision scenarios, but they do not describe all elastic collisions

C is incorrect because this describes an inelastic collision

4

The correct answer is **C** because:

- An elastic collision is a collision in which kinetic energy is conserved
- So conversely, an inelastic collision is a collision in which kinetic energy is **not** conserved
- Momentum is conserved in both elastic and inelastic collisions
- The principle of conservation of momentum states that:
  - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system

A is incorrect because the definition of an inelastic collision is that kinetic energy is not conserved as energy may be lost as sound, heat, light etc

B & D are incorrect because for any system where there are no external forces acting, momentum is always conserved

5

The correct answer is **A** because:

- The unit of density $\left(\rho = \frac{m}{v}\right)$ is $kg\ m^{-3}$
- The unit of volume is m^3
- The unit of velocity is $m\ s^{-1}$
- So, the unit of density × volume × velocity is
  - $[kg\ m^{-3}][m^3][m\ s^{-1}]$
  - $[kg][m\ s^{-1}]$

This is the same unit as momentum $(p = mv)$, which is $kg\ m\ s^{-1}$

	the unit of density $\left(\rho = \frac{m}{v}\right)$ is $kg\ m^{-3}$

B is incorrect because	the unit of energy $\left(E_k = \frac{1}{2}mv^2\right)$ *is* $J = kg\ m^2s^{-2}$ so, the unit of density × energy is $[kg\ m^{-3}][kg\ m^2s^{-2}] = kg^2\ m^{-1}s^{-2}$
C is incorrect because	the unit of pressure $\left(P = \frac{F}{A}\right)$ *is* $N\ m^{-2} = kg\ m^{-1}s^{-2}$ the unit of area is m^2 so, the unit of pressure × area is $[kg\ m^{-1}\ s^{-2}][m^2] = kg\ m\ s^{-2}$ (which is equal to N)
D is incorrect because	the unit of weight is $N = kg\ m\ s^{-2}$ the unit of area is m^2 so, the unit of weight ÷ area is $[kg\ m\ s^{-2}]/[m^2] = kg\ m^{-1}s^{-2}$ (which is equal to Pa)

6

The correct answer is **D** because:

- The principle of conservation of momentum states that:
  - For a system of interacting bodies, the total momentum remains constant provided there is no resultant force acting on the system
- In other words, in a **closed** or **isolated** system, the total momentum of two bodies before a collision is equal to their total momentum after the collision

A is incorrect because this is true, however it needs to clarify that momentum is only constant in a closed or isolated system

B & C are incorrect because these are definitions of force and momentum

7

The correct answer is **B** because:

- Kinetic energy = $\frac{1}{2}mv^2$
- Each sphere moves with speed v before the impact, so the kinetic energy of each sphere = $\frac{1}{2}mv^2$
- So, the sum of the kinetic energies is = $\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = mv^2$
- The spheres collide elastically, meaning no kinetic energy is lost so the kinetic energy before the impact = kinetic energy after the impact = mv^2

A is incorrect because this is the definition of a perfectly inelastic collision

C is incorrect because the total kinetic energy before impact would be zero if energy was a vector quantity, but it is a scalar so the total kinetic energy is the sum of the energies

D is incorrect because momentum is a vector, because the spheres are moving in opposite directions the total momentum is equal to mv – mv = 0

8

The correct answer is **D** because:

- A collision in which the objects stick together and have the same final velocity is called a perfectly inelastic collision
- This is because it reduces the initial kinetic energy more than any other type of inelastic collision
  - Option D is therefore correct because we are told the trolleys "become attached and move off together" indicating it is an inelastic collision and some kinetic energy is lost as a result
- The principle of conservation of momentum states that (in a closed system) the total momentum of the two trolleys before the collision is equal to their total momentum after the collision
  - Option D is therefore correct because there are no other bodies involved apart from the trolleys, so momentum must be conserved

A & B are incorrect because momentum cannot be "changed" into kinetic energy, and vice versa

C is incorrect because momentum cannot be "lost" via heat, as heat is a form of energy

- So, the sum of the kinetic energies is = $\frac{1}{2}mv^2 + \frac{1}{2}mv^2 = mv^2$
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A & B are incorrect because momentum cannot be "changed" into kinetic energy, and vice versa

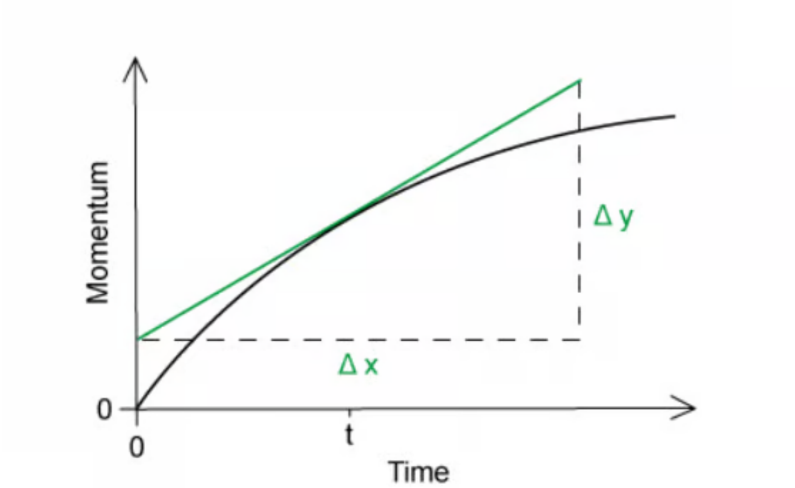
C is incorrect because momentum cannot be "lost" via heat, as heat is a form of energy and momentum is conserved in isolated systems

9

The correct answer is **B** because:

- The gradient of a momentum-time graph can be found at a particular time, *t*, by

drawing a tangent on the curve at *t* then using
  - $\frac{\Delta y}{\Delta x} = \frac{\Delta p}{\Delta t}$
- Force is equal to the rate of change of momentum so the gradient is equal to the resultant force on the car



A is incorrect because momentum is equal to p = mv, so velocity would be the gradient of a momentum-mass graph

C is incorrect because kinetic energy is equal to $E_k = \frac{1}{2}mv^2 = \frac{1}{2}pv$, so kinetic energy would be the area under a momentum-velocity graph

D is incorrect because the rate of change of kinetic energy is equal to power

10

The correct answer is **D** because the principle of conservation of momentum tells us that the momentum of a closed system is always conserved (provided no external forces act)

A & B are incorrect because it isn't possible to say that kinetic energy will **always** be conserved in a collision because the energy can be transferred to heat, light or sound

C is incorrect because when the tennis ball is struck by the racket, the velocity of the ball will increase, so the momentum of the individual ball will increase but the momentum of the racket will decrease to ensure the momentum of the system is conserved